TransformCare

Centers for Medicare & Medicaid Services, Assistant Secretary for Technology Policy Office of the National Coordinator for Health Information Technology

Health Technology Ecosystem

Response to: CMS-0042-NC

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1. Introduction

TransformCare, Inc., a Small Business Administration (SBA)-certified Service-Disabled Veteran-Owned Small Business (SDVOSB), is pleased to submit a response to the Centers for Medicare & Medicaid Services (CMS) Request for Information (RFI) CMS-0042-NC. Founded in 2017 by Dr. Kevin Dorrance, TransformCare is committed to revolutionizing healthcare delivery and optimizing outcomes and efficiencies for our federal, state, and commercial customers.

Our mission centers on advancing healthcare through transitioning to value-based, patient-centered care delivery models that empower consumers and improve outcomes. This response demonstrates how CMS is accelerating this transformation by enabling patient data ownership and supporting informed consumer choice in healthcare decisions.

From strategy development to implementation and sustainment, TransformCare's multidisciplinary team works collaboratively with clients to provide solutions to complex organizational issues. In this response, we highlight our expertise in managing large digital transformations and incorporate insights from our strategic partnerships, enabling Team TransformCare to comprehensively address CMS needs.

1.1 Company Profile

Since 2017, TransformCare has successfully executed more than twenty-five contracts across federal, state, and commercial healthcare sectors. Our work spans a diverse portfolio of initiatives aimed at translating research and health policy into real-world solutions that improve patient outcomes and generate cost savings for payors. The table below provides additional information relevant to contracting with TransformCare.

TransformCare Profile	
Company Headquarters	2751 Ridge Road Damascus, MD 20872
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CAGE Code	81TK8
Designation	SBA-certified SDVOSB
Contract Vehicles	Navy SEAPORT NxG GSA MAS (for discussion)

1.2 A Free-Market Healthcare System

The U.S. healthcare system continues to underperform despite spending $4.9 trillion annually, more than any other high-income country.[1] Billions of dollars are wasted yearly on administrative inefficiencies and antiquated payment models, while systemic complexities create opaque pricing, limit competition, and discourage individual

[1] "NHE Fact Sheet | CMS."

responsibility. In today's environment, patients lack meaningful choices, providers struggle under administrative burdens, and innovation stagnates under regulatory constraints.

Based on our experience and research, Team TransformCare believes transitioning to a free-market healthcare system provides the most appropriate and practical framework for prioritizing competition, informing consumer choice, and establishing appropriate regulatory boundaries to drive three primary objectives:

- Price competition and consumer choice
- Transparency in costs and quality
- Efficient and effective care delivery across virtual and physical settings

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The transition to a free-market healthcare system puts the patient at the center of value-based care.

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2. Patients and Caregivers

2.1 Patient Needs

2.1.1 Patient Empowerment Through Health Data Ownership

In a modern, patient-centered healthcare system, empowerment must begin with a patient's unambiguous ownership and control of their consumer electronic health record.

Drawing from the leadership team's decades of cumulative experience engaging with patients and caregivers, the Team TransformCare has consistently heard a unified message: Individuals want and deserve complete, continuous access to their comprehensive medical records. This access is essential not only for managing their own care but also for supporting vulnerable family members—elderly parents, family members with disabilities, or partners with chronic conditions.

Access to health information must not be viewed as a privilege but as a critical enabler of informed decision-making, continuity of care, and personal agency. In a free-market healthcare model, the ability to freely share health records with any provider, across any location or system, is a foundational requirement—not a secondary consideration.

CMS has the opportunity to lead by embedding data ownership, AI-enabled insights, and portability as core policy principles, ensuring every American can actively participate in their care journey with clarity, dignity, and choice.

2.1.2 Current System Barriers

Despite these clear consumer needs, current healthcare data fragmentation creates significant barriers to consumer choice and effective care management. Health information remains fragmented and siloed across multiple entities such as Electronic Health Record Systems (EHRs), insurance websites, pharmacy systems, and paper

records, preventing patients from functioning as informed consumers in a competitive marketplace. (PC-2, PC-2a)

Breaking down these barriers requires empowering patients with comprehensive tools and information that transform them from passive recipients of care into active, informed, and empowered healthcare consumers.

To function with personal agency in a competitive healthcare marketplace, consumers need six key capabilities: (PC-1a)

1. **Smart purchasing decisions** – AI-driven insights for selecting equipment, prescriptions, and services based on cost, quality, and personal health needs
2. **Provider selection** – Patient reviews, quality ratings, and AI-powered recommendations to choose the best providers.
3. **On-demand care access** – Immediate virtual visits, emergency mental health resources, and convenient care options.
4. **Personalized recommendations** – Tailored suggestions for products and services that address individual health conditions and goals.
5. **Health record queries** – The ability to ask questions about personal consumer electronic health records using advanced LLMs and natural language processing, powered by secure, private AI models.
6. **AI-powered health management –** Continuous monitoring and advanced machine-learning analysis of complete health records with new insights and recommendations, proactive reminders about medications and health issues, and complete access to audio, text transcripts, and summaries of all provider interactions—date-stamped and indexed for easy reference.

The accompanying video submission demonstrates how a consumer-controlled healthcare platform addresses the needs described above, providing Medicare beneficiaries and caregivers with guaranteed secure portability of all health records—a cornerstone of free-market healthcare. (PC-3) Click here to access our video: Health Technology Ecosystem - TransformCare Inc.

The current healthcare infrastructure imposes unnecessary burdens on patients seeking access to objective clinical data, such as X-rays, lab results, and diagnostic imaging. Outdated practices—such as imaging departments burning imaging data onto CD-ROMs for in-person pickup—persist across many systems. These methods not only rely on obsolete technology incompatible with most modern computers but also introduce multiple opportunities for delay, loss, or data corruption.

Even when patients do obtain their diagnostic data, results are often presented without interpretation or clinical context. This lack of clarity can lead to misunderstanding, unnecessary worry, and avoidable utilization of healthcare services.

To support a truly consumer-driven healthcare system, clinical information must be delivered in digitally accessible, context-rich, and actionable formats. This ensures patients can make informed decisions about their care in collaboration with providers, while improving system effectiveness and efficiency and reducing redundancy. (PC-2c)

Real-World Impact of a Fragmented Healthcare Infrastructure

A TransformCare Team member, a breast cancer survivor, relocated multiple times after her treatment. Despite faxing consent forms and repeatedly calling imaging centers to transfer her records, the original diagnostic images were permanently lost.

This loss is particularly concerning since longitudinal mammogram comparisons significantly improve diagnostic accuracy—a clear example of how data fragmentation undermines patient outcomes and consumer choice in healthcare markets.

2.1.3 AI-Powered Consumer Platform for Healthcare Decision-Making

Current digital health apps and legacy health data platforms lack the sophisticated capabilities needed for a competitive healthcare marketplace: contextual awareness, natural language processing for on-demand queries, and advanced machine learning that can reason across complete health records. Building on the six essential capabilities outlined above, CMS can control costs and put consumers firmly in control of their healthcare journey by combining universally accessible electronic health records with AI-powered analytics and intelligent digital assistants. (PC-4)

The next generation of healthcare apps and data platforms must be built around individual patients, providing on-demand access to their consolidated, evolving consumer electronic health records. This requires a fundamentally different approach to organizing health information. This app would “sit on top” of a completely reconfigured consumer electronic health record platform, representing the totality of a patient’s health information in an advanced, “ontology-governed,” personal clinical knowledge graph architecture. This unique approach connects all aspects of a patient's health data in meaningful multidimensional relationships, similar to creating a comprehensive map of one’s medical journey that AI can understand, navigate, and analyze. (PC-4a)

Utilizing new and novel health information representation architecture (ontology-governed, personal clinical knowledge graphs), over which advanced, deterministic, machine learning algorithms and fine-tuned proprietary large language models (LLMs) can infer and reason, CMS can finally answer the universal questions ALL patients ask: "*What are the best treatment options for people like me with my specific conditions*?" and "*What works for whom, why, how, when—and what are the tradeoffs*?"

This comprehensive consumer-controlled platform establishes the following key workflows: (PC-4b)

- **Competitive marketplace** – AI-generated insights and recommendations help consumers choose providers and services based on cost comparisons, quality ratings, and patient reviews
- **Proactive health coaching** – AI tools recommend lifestyle changes to help prevent the onset of adverse events or catastrophic illness
- **Risk-based care management** – Patient stratification into low, medium, or high-risk categories using proven tools, with AI monitoring clinical activity to reduce "rising-risk" patients through targeted interventions

2.1.4 Ensuring Universal Access and Usability

These AI-powered workflows must be accessible to all Medicare beneficiaries and caregivers, including those with limited digital experience. Essential design features for widespread adoption include: (PC-6)

1. **User-Centered Design**
    - Frictionless, intuitive interface with large, clearly labeled buttons and icons
    - Plain language with minimal text and no medical jargon
    - Essential tasks and content prioritized on each screen
2. **Accessibility and Inclusion**
    - Adjustable text sizes and keyboard navigation support
    - Voice input capabilities and multi-language support
    - Phone-based customer support and offline data access
    - Printable documents and summaries for non-digital users
3. **Seamless Onboarding and Support**
    - Optional step-by-step tutorials with help icons and tooltips
    - Secure caregiver proxy access with clear privacy settings
    - Password-free login options (biometric, PIN, SMS codes)
    - Trusted symbols displaying Medicare and HIPAA compliance
4. **Actionable Health Management**
    - Voice-activated queries for hands-free information retrieval
    - Personalized dashboards with key health information
    - Automated reminders for medications and appointments
    - Simple charts and graphics explaining health trends
    - Quick links for everyday tasks like prescription refills
5. **Comprehensive Support**
    - Live chat and extended-hours phone support
    - Easy-to-follow video tutorials and printable guides

- Cross-device compatibility with mobile-first design
- Continuous improvement through user feedback and analytics

CMS has a unique opportunity to drive widespread adoption of consumer-centric digital health platforms through strategic marketing campaigns, targeted education programs, and incentives that disrupt the current fragmented system. (PC-5)

2.1.5 Platform Development and Oversight

CMS can ensure digital health platform quality by facilitating thorough and exhaustive discussion and dialogue between CMS and the App Platform Development teams, focusing on the foundational health information representation architecture, academic research fidelity and veracity, model flexibility, model scalability, model reproducibility, model sustainability, model robustness, cybersecurity infrastructure robustness, and compliance infrastructure. (PC-5a)

Increased patient and caregiver adoption of patient-centered digital health products will occur when these new tools and workflows can objectively demonstrate enhanced patient education, engagement, compliance, and personal agency working in service to scalable, reproducible, and sustainable Shared Decision Making.

Enabling timely access to high-quality patient data requires a new, novel, advanced, patient-level health information representation platform architecture: Ontologically Governed, Personal Clinical Knowledge Graphs. These architectural platforms are built to ingest previously siloed, fragmented, multi-modal, patient-level, structured, and unstructured data/information. The patient's health information is represented according to defined semantic (ontology) guidelines and structured for advanced analysis by machine deep learning algorithms. The patient's consumer electronic health record is treated as machine-readable "Code," and the novel representation architecture functions as a new patient-level "Operating System." (PC-5b)

This approach treats patient health information as structured knowledge that can be processed by sophisticated AI systems, functioning as a patient-centered "operating system" for healthcare decision-making.

To collect real-world data on health outcomes, consider incentivizing and/or mandating participation in real-world data collection as a condition of CMS reimbursement for new digital health technologies. This could include reporting patient outcomes, treatment/medication adherence, impact on downstream resource utilization (e.g., hospitalizations, ED visits, etc.), clinical end points goals (BP, weight, HbA1c, lipid management, etc.), readmission rates, patient-reported outcomes, device usage metrics, usability, satisfaction, and quality of life impacts. (PC-7)

2.1.6 The Patient-Centered Free Market Vision

These patient needs collectively point toward a fundamental transformation: empowering consumers to drive healthcare competition through data-informed choice. When patients own their data, access AI-powered insights, and shop in a transparent marketplace, they become the catalyst for system-wide improvement.

CMS has the opportunity to lead this transformation—moving from price-setter to platform enabler, putting patients firmly in control of their healthcare journey.

2.2 Data Access and Integration

2.2.1 From Fragmented Data to Actionable Knowledge

The health data most readily available and valuable to patients and caregivers include medications, diagnoses, lab results, and imaging reports, as well as comprehensive, interrelated visit summaries (PC-8). HIPAA guarantees individuals the right to access their complete medical records, and modern patient portals have made much of this information immediately accessible.

Despite this technical accessibility, a fundamental challenge persists—patient health data remains fragmented across providers, payers, and legacy systems (PC-8). Information is siloed and poorly integrated, lacking the functional and logical (contextual) interoperability needed to create meaningful, actionable insights for informed healthcare decisions.

The Current Reality

Current systems fail to deliver what consumers need for informed healthcare decisions, as patient information remains scattered across multiple touchpoints without proper integration or context (PC-8). Information that exists in isolation—a lab value here, a prescription there, a diagnosis code elsewhere—provides limited utility for understanding overall health status or making informed care decisions.

High-Value Data Categories

Beyond basic clinical records, social determinants of health (SDOH)—such as housing status, food security, transportation access, education, and employment—represent some of the highest-value data sources for comprehensive patient understanding (PC-8b). Additionally, patient-reported outcomes, functional assessments, and behavioral health indicators provide crucial context that traditional clinical data often misses.

The Path Forward

The transformation requires moving from data availability to data usability—creating integrated, contextualized information that empowers patients and caregivers to make truly informed healthcare decisions (PC-8). This means not just providing access to data but delivering actionable knowledge that connects individual health information to broader health outcomes and care options.

2.2.2 Patient Data Ownership: The Free-Market Foundation

In a free-market healthcare system, patients—not providers, payers, or EHR vendors—own their consumer electronic health record. The most valuable yet hardest-to-access information is the Continuity of Care Document (C-CDA), containing the complete clinical picture: encounter details, procedures, medical history, medications, allergies, problem

lists, lab results, and clinical notes. Patient access to this comprehensive data must be immediate and unrestricted. (PC-8a, 8b)

Team TransformCare's platform demonstrates this principle in action, leveraging national exchanges and state Health Information Exchanges (HIEs) to connect over 125,000 clinical endpoints, delivering comprehensive C-CDA records directly to patients and their chosen care teams. (PC-8)

2.2.3 Beyond Clinical Data: The Complete Patient Picture

The highest-value data sources extend beyond clinical records to include social determinants of health (SDOH)—housing status, food security, transportation access, education, and employment. These factors directly impact care adherence, recovery, and outcomes. Combined with real-time Admission, Discharge, and Transfer (ADT) feeds, this comprehensive data enables personalized care and precise interventions that support informed consumer choice. (PC-8b)

2.2.4 Transforming Data Architecture for Consumer Empowerment

True patient empowerment requires sophisticated data representation architecture that goes beyond simple connectivity. Many assume that structured data automatically provides meaning, but the confusion between structural interoperability (how data is stored), logical interoperability (how data is understood), and knowledge management (how knowledge is captured, and information is derived and utilized) leads to major challenges in health and health-related data exchange, integration, and reasoning. There is important interplay between structure and semantics. It is important to know why databases are simply containers (current EHRs), knowledge bases (graphs) capture information, and knowledge management systems support reasoning and decision-making.

2.3 Interoperability and Clinical Research Integration

Improving accessibility, interoperability, and integration of clinical data requires a foundational shift in how we structure and interpret health information. The modern healthcare digital landscape is tightly interconnected, yet interoperability and knowledge management are often discussed but not always well understood. (PC-8c).

2.3.1 Key Challenges

Information Blocking and Regulatory Barriers: Despite federal regulations like the 21st Century Cures Act, information blocking remains a significant challenge. Healthcare organizations sometimes resist data sharing due to competitive concerns, technical limitations, or compliance uncertainties.

Technical and Structural Limitations: Much of today's health data exists in unstructured formats or proprietary systems that resist integration. Many assume that structured data automatically equals interoperable data, but true interoperability requires semantic understanding—not just technical connectivity. Legacy systems often lack modern APIs,

and even FHIR-compliant systems may interpret data standards differently. Current legacy EHR platforms demonstrate no operational logical data interoperability.

Data Quality and Standardization: Inconsistent data quality, varying terminologies, and lack of standardized coding across systems create barriers to meaningful integration. Clinical data often lacks the context and metadata necessary for research applications.

2.3.2 Key Opportunities

Advanced Standards Implementation: Widespread adoption of FHIR R4 and other modern interoperability standards creates opportunities for real-time data exchange and integration. When properly implemented, these standards enable seamless data flow between previously incompatible systems.

AI-Powered Data Integration: Machine learning and natural language processing technologies can help extract meaning from unstructured clinical notes, radiology reports, and other text-heavy data sources, making previously inaccessible information available for enhanced patient-level insights as well as clinically relevant research.

Patient-Mediated Information Exchange: Empowering patients to control and direct their health data creates new pathways for data integration across care settings. Patient-directed exchange can overcome some institutional barriers to data sharing.

Real-World Evidence Generation: Integrated clinical data enables more robust real-world evidence studies, supporting regulatory decision-making and clinical guideline development. The combination of claims data, clinical records, and patient-reported outcomes creates a more complete patient-level picture for both clinical and operational outcomes improvement and advanced real-world research.

2.3.3 Strategic Recommendations

1. **Semantic Interoperability Focus:** Move beyond technical connectivity to ensure data maintains meaning and context across systems.
2. **Knowledge Management Systems:** Implement platforms that do not just store data but create actionable clinical knowledge.
3. **Privacy-Preserving Technologies:** Leverage technologies like federated learning and differential privacy to enable research while protecting patient confidentiality.
4. **Stakeholder Alignment:** Create incentive structures that encourage data sharing while addressing legitimate competitive and privacy concerns.

The ultimate opportunity lies in creating a healthcare ecosystem where clinical data flows seamlessly to support both individual patient care and population health management, generating evidence that continuously improves care quality and outcomes across the entire healthcare system.

2.3.4 Federal Program Limitations and Market Solutions

Despite efforts to improve healthcare data access, current federal initiatives fall short of enabling true consumer choice in healthcare markets. Since 2010, federal programs including Meaningful Use, the 21st Century Cures Act, and TEFCA have aimed to improve

healthcare data interoperability. While well-intentioned, these initiatives have introduced complexity and unintended consequences that actually hinder consumer empowerment and market competition. (PC-9, PC-10)

Key Implementation Barriers (PC-8c)

- **Information Blocking** – Health systems delay or deny data access despite federal regulations
- **Cost & Complexity** – Data integration requires significant resources, especially for smaller organizations
- **Inconsistent Standards** – EHR vendors code data differently, requiring extensive normalization

TEFCA's Limited Impact

TEFCA's voluntary participation has resulted in minimal adoption—only 2,816 organizations as of late 2023, representing a fraction of America's 6,000+ hospitals and hundreds of thousands of clinics. This stems from its voluntary nature, mutual participation requirements, high implementation costs, and complex legal agreements.

Consumer Impact

Patients still cannot access complete medical records across systems, leading to redundant testing, higher costs, fragmented care, and reduced coordination effectiveness—precisely the opposite of what free-market healthcare requires.

Expanding Data Access

To enable true consumer choice in healthcare markets, CMS should consider expanding data sources beyond Blue Button 2.0 to include Medicare Advantage encounter data, clinical EHR data, and social determinants of health information. However, additional barriers prevent effective access including data silos with incompatible systems, privacy compliance burdens, and resource constraints limiting smaller organizations' integration capabilities. (PC-9a)

2.3.5 The Consumer-Centric Solution

The most effective path forward empowers patients as the sole source of truth for their medical records. When data is aggregated, normalized, analyzed, and operationalized for patient use, information becomes shareable, portable, and actionable, enabling true interoperability, information dominance, and decision advantage that serves consumer choice rather than system convenience.

2.3.6 Emerging Use Cases: Transforming Healthcare Through Consumer Intelligence

Referencing the advanced health information representation architecture and scaling this opportunity across CMS populations, multiple high-value use cases emerge that directly support transformative patient-centered care: (PC-12a – 12d)

1. Substantial enhancement of patient-level understanding, education, engagement, compliance, personal agency.

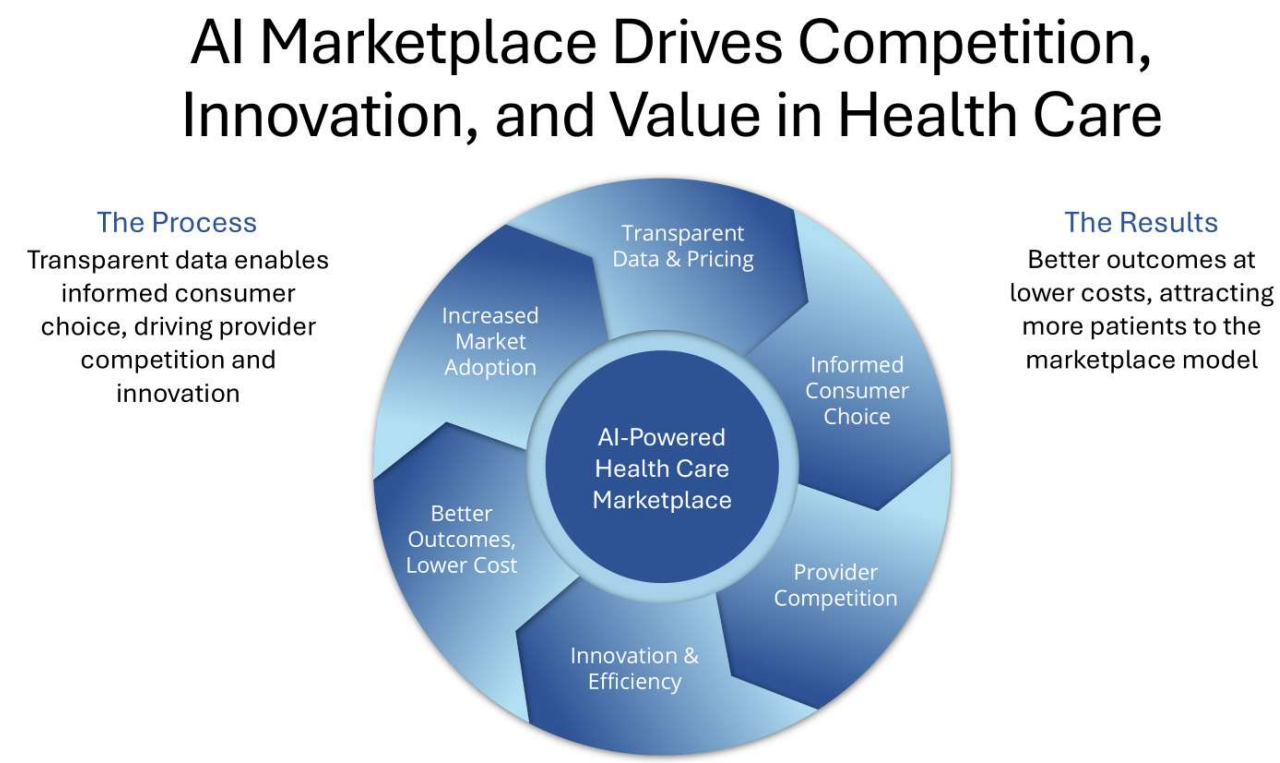
2. A legitimate, objective, scalable, reproducible, and sustainable roadmap to Shared Decision Making.
3. A "real world", "information-driven" mechanism to identify and characterize root causes of "Needless Care Variation."
4. Given the accepted presence of "Needless Care Variation," and therefore the presence of "Low Value Care," this new knowledge management system will have the computational power and granularity to identify and stratify across the spectrum from "Low Value Care" to "Best-in-Class Care" for any given and highly specific medical condition occurring in highly similar patient cohorts. If it is agreed that "Low Value Care" exists and is identifiable, then "Best-in-Class Care" must also exist and be identifiable under specific constraints. If this is true, then "Best-in-Class Care" pathways, recommendations, and guidance could be widely "Democratized" for the benefit of all.
5. A mechanism to answer universal patient questions/desires: "What is the best treatment advice and treatment journey for people like me who have the same medical condition that I have?" and "What works for whom (why, when, how) and what are the tradeoffs?"
6. A "real world", "information-driven" population health management and value-based care knowledge management platform.
7. Improved patient outcomes.
8. Improved patient safety.
9. Improved operational efficiency.
10. Improved mitigation of healthcare inequities, disparities, and biases.
11. **The emergence of an AI-powered intelligent EHR platform that creates scalable, reproducible, and sustainable value for patients, providers, and payers.**

These transformative capabilities become possible through the advanced knowledge management infrastructure described above, where ontology-governed personal clinical knowledge graphs represent multi-modal raw data. The raw data is converted to functional "Code" within the knowledge graph, which can then be analyzed by deterministic machine learning algorithms and reasoned over and queried by fine-tuned LLMs. This process leads to "Data Operationalization," which leads to "Information Dominance," which leads to "Decision Advantage." This value creation accrues to the patients: **This is the massive opportunity!**

2.3.7 The Market Impact

This AI-powered platform creates a positive feedback loop that drives continuous healthcare improvement, as illustrated in Figure 1.

Figure 1: The Free-Market Healthcare System



The result is scalable, reproducible, and sustainable value creation for patients, providers, and payers—the foundation of successful free-market healthcare.

2.4 Information Blocking and Digital Identity

An increase in reported information blocking incidents can serve as a critical mechanism for identifying and addressing entities that hinder the exchange of health information. Early identification of such actors enables timely course correction and promotes broader compliance with federal interoperability standards.

To support this effort, CMS could strengthen patient awareness and engagement by incorporating clear, patient-facing prompts that encourage the reporting of information blocking. These could be included in:

- Explanation of Benefits (EOB) statements
- Medicare handbooks or enrollment materials
- Digital platforms such as embedded links to the ONC complaint portal within Blue Button 2.0 and the MyMedicare mobile app

Digital identity credentials and associated credentialing service providers (CSPs) are useful for safe health data exchange. Yet, the healthcare industry faces challenges with

utilization and adoption. Most patients/caregivers have never heard of NIST or digital identification systems. There is little or no awareness of these systems' benefits and separate logins. Since patients/caregivers are required to recall existing logins for local portals, patients and caregivers see little payoff. (PC-14-a)

If consumers used single sign-on more frequently, the benefits would be clear. These logins allow an independent platform to house medical records and cost and quality data, access data, and permit real-time comparison-shopping. Every credentialed provider can pull participating patients' complete longitudinal records. It would also enable family proxies to manage care without the traditional paperwork required, speeding up market demand for home-based preventative services. The downsides include potential privacy concerns, credential recovery hassles that could delay urgent care, and potential exclusion risk for tech-limited patients. However, these risks or challenges are no different than the contemporary patient portals and EMRs. (PC-14b,PC-14c)

CSPs improve access by giving any consumer app a pathway to universal healthcare data. CSPs map cleanly to individual patients in any EHR, reducing duplicate and mismatched records. This credentialing allows patients to grant access to particular providers more easily. If providers were to model adoption and switch to CSPs for their portal, usage would increase. CMS could drive adoption by incentivizing early adopters through free marketing on their portals and interfaces. (14d,14e)

3. Providers

3.1 Digital Health Apps

3.1.1 Provider Adoption Strategy

Based on Team TransformCare's experience implementing digital health transformations, CMS should focus on directly empowering Medicare beneficiaries to adopt digital health products that support them and their care teams, rather than encouraging providers to adopt these tools. When patients are equipped with the right tools and access, they become active participants in their care—driving adoption, improving outcomes, and reducing the burden on the system.

3.1.2 Building Trust Through Technical Excellence

The key to successful adoption is engendering unmitigated TRUST in specific digital health products and data platforms. This trust is achieved, supported, and maintained through several critical capabilities: (PR-1)

1. **Data Foundation and Access**
    - Quality, veracity, and fidelity of the foundational, patient-specific data on which the product is built
    - Effective and efficient access and use of patient-level data through superior UX/UI functionality

2. **Technical Transparency**
    - A general understanding of the foundational data representation architecture (ideally, ontology-governed personal clinical knowledge graphs)
    - A general understanding of the machine learning models, reasoning engines, and LLMs that comprise the backend technology stack
3. **Clinical Decision Support**
    - Functionality that facilitates, verifies, and memorializes shared decision-making efforts
    - Functionality that delivers on-demand data analysis and operationalization capabilities
4. **Value-Based Care Enhancement**
    - Functionality that identifies and promotes real-world "high-value care," thereby mitigating low-value care, needless care variation, disparities, inequities, and biases
    - Functionality that facilitates precision medicine derived from advanced knowledge management systems

When these trust-building elements are in place, patients become confident advocates for digital health adoption, naturally driving provider engagement through demonstrated value and outcomes.

However, to equip patients with the right tools and access to approved digital health products, CMS must first address the fundamental challenges described below. (PR-1a)

Fundamental Data Architecture Limitations

On-demand data analysis and operationalization are unachievable with the current legacy EHR data representation architecture, preventing the real-time insights patients need for informed decision-making.

Rural Healthcare Barriers

- **Financial Constraints** – Rural healthcare providers often operate with limited budgets, making it challenging to invest in new digital health technologies and the necessary infrastructure.
- **Workforce Shortages** – Rural areas experience a scarcity of healthcare professionals, which can impede the implementation and maintenance of digital health solutions.
- **Technical Challenges** – Many rural providers lack the technical support and expertise required to integrate digital health products into their existing systems.
- **Regulatory and Reimbursement Issues** – Complex regulations and uncertain reimbursement policies can deter providers from adopting digital health technologies.

Provider Communication Guidelines

When using digital health products in the provision of care, providers should share the following information with patients: (PR-1b)

Purpose and Benefits

Explain that the tool helps patients aggregate their medical records, access them in one place, and share with any provider or care partner when needed. Emphasize improved care coordination, reduced duplicative testing, and enhanced safety through full health record visibility.

1. **Data Access and Use:**
   - Clarify that the patient's records are collected from multiple clinical sources, securely stored, and only shared with their explicit permission.
   - Describe how this data helps both patients and their care teams make more informed decisions.
2. **Patient Rights and Control:**
   - Reinforce that the patient is the sole owner and controller of their health data.
   - Remind patients that they can view, manage, and share their information with any provider—on their terms.
3. **Security and Privacy:**
   - Communicate that the platform uses HIPAA-compliant infrastructure and secure direct messaging to protect health data at all times.
4. **Support and Guidance:**
   - Let patients know that help is available for setup, navigation, and troubleshooting, ensuring a smooth and empowering experience.
   - This approach builds trust and reinforces the core mission: giving individuals ownership and control of their health information to drive better outcomes and a more connected care journey.

Provider Responsibilities

Providers have the responsibility to build **trust**—performance, intent, and integrity—in the digital health product platform. When recommending the use of digital health products, providers have several key responsibilities: (PR-1c)

1. **Clinical Appropriateness:**
   - Ensure the product is relevant to the patient's condition, care goals, and digital literacy level.
2. **Informed Guidance:**
   - Clearly explain how the product works, what it does, and how it supports the patient's care.
   - Set realistic expectations for outcomes and engagement.

3. **Privacy and Security Disclosure:**
    - Inform patients about how their data will be used, stored, and protected.
    - Confirm that the product complies with HIPAA and other applicable regulations.
4. **Patient Empowerment:**
    - Reinforce that the patient maintains control over their data, especially in consumer-driven models.
    - Support the patient's right to access and share their information as they choose.
5. **Ongoing Support:**
    - Offer help in getting started with the product and provide follow-up for questions or concerns.
    - Monitor the product's role in the care plan to ensure it is adding value.

Providers play a crucial role in ensuring digital health tools are used safely, ethically, and effectively to improve patient outcomes.

3.1.3 Obstacles to Innovative Application Development

Current commercial EHR platforms have inadequate data/knowledge representation architecture, therefore preventing time-sensitive, context-aware, point-of-care, and point-of-task information dominance and decision advantage that would otherwise support physician workflows. (PR-2)

Ontology-governed, personal clinical knowledge graphs achieve what current EHR platforms cannot.

3.1.4 Technical Challenges and Solutions for Full Data Access

It is critical for healthcare delivery—both in urban and rural areas—that all health data, regardless of the EHR system or storage format, is available and accessible for exchange. However, true accessibility goes beyond system-to-system exchange. (PR-3)

When patients are in control of their data, they can collect, synthesize, and present a complete, unified medical record to any provider at any point of care. This approach ensures:

- No gaps in medical history, regardless of geography or EHR vendor.
- Better care coordination and decision-making.
- Fewer duplicate tests and avoidable errors.
- A consistent standard of care, whether in a rural clinic or a major urban hospital.

Patient-controlled access breaks down the silos, allowing healthcare to follow the individual—not the system.

Current Data Format Challenges

Current challenges in accessing different data formats stem from the inconsistent way EHR systems structure and code clinical data. Even when using the same standards (like

CCDs or C-CDAs), each system can apply its own formatting, terminology, and schema variations—making true interoperability extremely difficult. (PR-3a)

This fragmentation is a systemic issue that prevents seamless data exchange. To overcome it, there must be access to data normalization engines that can ingest, interpret, and consolidate various formats into a single, cohesive, and transferable medical record. Without normalization, different data sources cannot effectively "speak the same language," and the burden of that gap often falls on patients and care teams.

Impact on Care Quality

The lack of standardized data formats and the absence of effective normalization have a direct and negative impact on patient care quality. When data is fragmented or inconsistent across EHR systems: (PR-3b)

- Providers lack a full picture of the patient's medical history, increasing the risk of misdiagnosis, medication errors, or redundant testing.
- Care coordination suffers, especially during transitions between hospitals, specialists, or across state lines.
- Treatment delays occur due to time spent reconciling or requesting missing information.
- Most importantly, patients are left to fill in the gaps, placing the burden on them to manage complex and incomplete records.

High-quality care depends on complete, accurate, and timely information. Without data normalization and patient-centered aggregation, that standard is nearly impossible to achieve.

Technical Barriers to Full Data Accessibility

The technical barriers to full data accessibility are rooted in the inconsistent formatting, storage, and structuring of data across EHR systems. Key challenges include: (PR-3c)

1. **Lack of Standardization** – Even with common formats like C-CDA or FHIR, data is often coded and structured differently across vendors, requiring custom integration efforts.
2. **Schema Variability** – Each EHR may label, categorize, or organize clinical concepts (e.g., medications and diagnoses) in unique ways, making direct comparison or aggregation unreliable.
3. **Limited Interoperability** Infrastructure – Many systems lack the APIs, endpoints, or exchange mechanisms necessary to export and import comprehensive patient records across platforms.
4. **Absence of Normalization Engines** – Without tools to standardize, map, and reconcile disparate data sources, it is impossible to generate a unified, patient-accessible record that can travel across the care continuum.
5. **Data Silos and Proprietary Lock-In** – Some EHRs or health systems restrict access, making it difficult for third-party platforms or patients to extract complete data sets.

Without robust normalization and patient-centered data design, full accessibility remains a fragmented and inefficient goal.

3.1.5 The Patient-Centered Solution

Based on established health utility principles, expecting all healthcare entities to fully invest in and align around true interoperability is a significant challenge—both technically and politically. Every system has its own priorities, budgets, and incentives, which makes universal alignment slow and expensive. (PR-3d, PR-3e))

That is why putting the patient at the center—leveraging their HIPAA-granted right of access—is the most scalable and cost-effective solution. When data flows through the patient, the need for massive system-to-system integrations is reduced. Instead, data can be aggregated, normalized, and made interoperable at the point of patient access, dramatically lowering costs, increasing portability, and giving individuals full control over their health information. This approach transforms a complex infrastructure problem into a human-centered solution.

Unlike traditional system-to-system models—slowed by fragmentation, inconsistent data formats, and misaligned incentives—patient-driven access bypasses institutional barriers, reduces costs, and accelerates real-world impact. It empowers patients as the constant across care settings, ensuring: (PR-3e)

- Continuity of care, even across disconnected systems or rural/urban divides
- Data normalization at the point of access, not within every silo
- Improved safety, advocacy, and outcomes through transparency and control

Patient-centered data interoperability is foundational to healthcare transformation. While system-level interoperability remains important, the patient-centered approach offers the most actionable, scalable path to solving the broader challenge.

3.1.6 Standards or Policies for Access to Third-Party Products

To enable patients and third-party digital products to access administrative workflows, CMS needs: (PR-4)

- Standardized, universal intake forms that are accepted nationally across providers and systems.
- Open, secure APIs that allow third-party tools to view provider directories, schedules, and availability.
- Clear policy guidance ensuring patients can delegate access to manage appointments and records through trusted apps.
- EHR scheduling integration, which remains a challenge due to proprietary systems and varied provider workflows—this requires vendor cooperation and incentives.

Without these policy changes, administrative access will remain fragmented and limit patient-driven interoperability

3.2 Data Exchange

3.2.1 Platform Capabilities and FHIR Integration

From the perspective of our platform, which includes a fully integrated FHIR server and standardized APIs, Team TransformCare is uniquely positioned to support seamless data exchange. As a consumer health utility, we aggregate the individual's comprehensive Continuity of Care Document (C-CDA) at a national level, regardless of where the care was delivered. (PR-5 – PR-5i)

Our platform can then convert that data into structured FHIR resources—such as Conditions, Medications, Allergies, and Encounters—ready for distribution in the format preferred by the receiving system or application. This bridges the gap between legacy document-based exchange and modern API-first interoperability, all while keeping the patient in full control of their data.

3.2.2 TEFCA Assessment and Limitations

TEFCA is not currently helping providers meaningfully advance access to health information. Despite good intentions, its low participation rates, technical complexity, and lack of real-world adoption mean most providers see no measurable benefit. There simply is not enough incentive or operational support for physicians—especially those bound by restrictive EHR and health system policies—to go outside their workflow to retrieve additional data, even when it would help their patients. (PR-6-6d)

3.2.3 Patient-Centered Data Exchange Strategy

CMS and its partners can simplify clinical quality data responsibilities for providers by removing the burden of data stewardship and allowing them to focus on delivering care. The best way to do this is by shifting to a patient-centered interoperability model, where patients collect, manage, and share their own health data across systems. (PR-8)

This strategy reduces administrative overhead, improves data accuracy, and ensures providers have the right information—without making them responsible for chasing or maintaining fragmented records. Simply put, care teams should provide care, not manage interoperability.

Bulk FHIR Data Export Considerations

Benefits: Using bulk FHIR data exports from EHRs to CMS could help streamline clinical quality reporting, reduce manual entry, and enable faster, standardized submissions at scale.

Challenges: Not all provider practices—especially smaller or downstream ones—are fluent in FHIR, making it difficult to interpret or act on that data. Additionally, FHIR resources can lack the full clinical context needed for quality improvement.

Recommended Solution: A comprehensive C-CDA, driven through patient-centered interoperability, offers a more complete view of the patient's healthcare journey. Empowering patients to collect and share their full record improves data completeness

and allows providers to make more meaningful, outcome-driven care decisions—regardless of technical fluency.

3.2.4 Consolidated Interoperability and Quality Reporting

Interoperability and quality reporting responsibilities can be consolidated by removing the burden from providers and shifting it to a centralized, patient-centered platform. Providers should focus on delivering care—not managing fragmented data systems. (PR-8c)

By investing in a consumer health utility model, all data—clinical, claims, and patient-generated—can be aggregated, normalized, and reported through a single platform. This approach streamlines compliance, improves data accuracy, and makes better use of existing investments by centralizing both interoperability and quality reporting in one unified system.

3.3 Digital Identity

CMS can encourage providers to accept patient-issued digital identities and credentials by promoting a model where the patient is the central, trusted source of truth—not the individual provider systems. (PR-9)

Using a consolidated interface and infrastructure, patients can manage and share their verified identity and clinical information from one secure platform, eliminating the need for providers to track multiple proprietary logins.

This simplifies access, reduces friction, and shifts the focus to the only relationship that truly matters: the one between the patient and their care team. CMS should support standards and policies that recognize patient-controlled credentials as valid and interoperable across all provider systems.

(No responses for: PR-10 – PR-14)

4. Payers

In a consumer-driven healthcare marketplace, payers play a critical role as facilitators rather than gatekeepers, enabling patient choice through transparent data sharing and streamlined processes. Current payer systems often create barriers that prevent patients from accessing their data and making informed healthcare decisions. Transforming payer operations to support consumer empowerment requires addressing technical limitations, improving data interoperability, and aligning incentives with patient-centered care.

4.1 TEFCA Limitations and Market Solutions

TEFCA faces several challenges that hinder payer participation and limit consumer empowerment. First, it offers few well-defined, payer-specific use cases—such as prior authorization, real-time eligibility checks, or claims adjudication—making the investment uncertain. Second, becoming a Qualified Health Information Network (QHIN) or sub-participant is complex and costly, especially for smaller or mid-sized payers. Third, variations in state and local privacy laws, patient consent requirements, and trust

frameworks make national standardization difficult. Fourth, patient matching inconsistencies undermine confidence in reliable, cross-network data exchange. (PA-1)

To address these limitations, TEFCA should provide clearer payer use cases and detailed playbooks on technical and legal requirements. Financial or technical assistance, such as grants or pre-approved agreements, would help smaller entities onboard. A consistent, standardized approach to patient identity and consent would ease cross-state data sharing. Without these improvements, smaller organizations may adopt a wait-and-see stance, slowing widespread TEFCA participation and limiting consumer access to comprehensive health data.

4.2 API Adoption and Consumer Empowerment

CMS can encourage payers to implement APIs similar to Blue Button 2.0 and Data at the Point of Care by linking adoption to Medicare Advantage and Medicaid managed care requirements. Offering financial incentives or tying higher Star Ratings to FHIR-based API capabilities can stimulate investment. Clear technical guidance, including reference implementation toolkits and publicly available sandbox environments, can reduce development complexity. Collaborative pilot programs with EHR vendors and provider groups can demonstrate the real-world efficacy and return on investment of data-sharing APIs, helping overcome organizational inertia. (PA-2)

4.2.1 Digital Identity Integration

Encouraging payers to accept digital identity credentials rather than proprietary logins could involve leveraging federal assurance standards (e.g., NIST 800-63) and requiring or incentivizing payers in Medicare Advantage or Medicaid managed care contracts to adopt these solutions. Consumers benefit from reduced login complexity and increased security, enabling seamless access to health data across multiple platforms. CMS could complement these efforts by running public education campaigns promoting standardized digital identities and working with ONC and other agencies to harmonize identity assurance efforts across the healthcare ecosystem. (PA-3)

4.3 Infrastructure Improvements for Market Efficiency

4.3.1 National Provider Directory

A robust national provider directory that includes FHIR endpoints and digital identity credentials would be highly valuable for both payers and consumers. Such a directory would simplify locating up-to-date provider information—practice locations, network affiliations, and secure endpoints—streamlining workflows like prior authorization requests, real-time eligibility checks, and care coordination. Creating a single source of truth would reduce duplicative maintenance activities, cut administrative costs, and improve data accuracy across the industry, ultimately enabling better consumer choice through accurate provider information. (PA-4)

4.3.2 Unified Knowledge Management Architecture for the AI Agent Economy

In a "Perfect World" and the context of "First Principles," each of the three major stakeholders occupying the center of the healthcare ecosystem (patients, providers, and

payers) would have access to the same knowledge management system. This patient-centered, ontology-governed, personal clinical knowledge graph (PCKG) data architecture would function as the single source of truth for patient, provider, and payer. It is a living, constantly evolving, and persistently updated digital, multimodal data representation of the patient that offers multiple unique value propositions for each stakeholder.

When multimodal, patient-level, health and health-related data is represented as proposed, the patient's complete longitudinal healthcare journey is modeled in an interoperable (logical), structured digital format that is ready for advanced AI-supported (machine learning) analysis. Now and only now, can complex interoperability and quality reporting questions be answered in an affirmative, time-sensitive, context-aware, stakeholder-specific, scalable, reproducible, sustainable, transparent, explainable, and uniform fashion. (See Section 4.3.4 for detailed implementation of PA-5a,b solutions.)

This unified approach becomes even more critical as healthcare rapidly approaches an "AI Agent Economy" that will facilitate embedding human-level decision-making into autonomous software. These AI agents will contain semantic capability layers within the technology stack that allow them to interpret context, make decisions, and perform tasks with minimal human supervision. These multi-specialty agents will be able to listen, reason, decide, and act.

The upcoming AI Agent era requires a necessary and sufficient knowledge management system over which these agents can fully exploit their capabilities. The current EHR data architecture will not support these goals, making the unified knowledge management approach essential for future healthcare operations.

Issues of data governance, scope of data access, digital security, privacy compliance, digital-user monitoring, "read only" status, "read and write back to" status, platform ecosystem architecture, as well as other "systemic matters" require further thoughtful and vigilant consideration.

4.3.3 Quality Data Burden Reduction

Payers can help reduce providers' quality data burdens by aligning on standard measure sets and supporting common data models. Encouraging EHR vendors and providers to adopt FHIR-based data exchange, especially bulk data extraction, can eliminate manual abstraction and reporting that weigh heavily on providers. Many payers are interested in receiving large datasets directly from EHRs, allowing them to calculate quality measures themselves. Providers, particularly smaller practices, generally welcome offloading this responsibility, provided data privacy and workflow impacts are addressed. (PA-5)

Payers see great potential in EHR technology advances that enable bulk extraction of clinical quality data, as it offers consistent measure calculation across their networks. To consolidate interoperability and quality reporting requirements, CMS and payers should harmonize measure specifications, adopt shared reporting platforms, and align data submission timelines. By standardizing Electronic Clinical Quality Measure (eCQM) logic and using FHIR-based reporting tools, providers can report once while satisfying multiple payers' requirements. Payers that invest in real-time data capabilities—such as providing

dashboards or APIs back to the point of care—can help clinicians address quality gaps proactively, improving patient outcomes and reducing administrative overhead. (PA-5a, PA-5b)

4.4 Information Blocking and Compliance

CMS can motivate payers to report information blocking by providing clear, step-by-step guidance, confidentiality protections, and incentives for credible complaints. This approach discourages data withholding and fosters a culture of compliance with interoperability rules. While excessive or poorly guided complaints might create conflict and slow collaboration, a responsible complaint system can illuminate real barriers and advance seamless data exchange that serves consumer choice rather than administrative convenience. (PA-7)

The result is a healthcare ecosystem where payers compete on value and service rather than restricting access, ultimately benefiting patients through better outcomes, lower costs, and greater choice in their healthcare decisions.

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5. Technology Vendors, Data Providers, and Networks

5.1 Ecosystem

5.1.1 Core Strategy: Enable Patient-Centered Decision Making

CMS should support, facilitate, incentivize, and, where appropriate, mandate data-driven Shared Decision Making (SDM) tools. To truly place Medicare beneficiaries at the center of healthcare, they must be empowered with education, engagement, and decision-making authority from the beginning of their healthcare journey. (TD-1)

Currently, while SDM is often discussed with good intentions, there is no objective infrastructure platform that supports scalable, reproducible, and sustainable shared decision making. Treatment decisions must be heavily influenced by quality-of-life and outcome goals that are important to each individual patient.

5.1.2 CMS Data Expansion for Innovation

With the addition of OAuth access to Medicare claims (Parts A, B, D) for patient-facing apps, CMS's Blue Button 2.0 API provides the foundation for leveraging clinical and claims data to tailor AI agents' conversations to each patient. However, the current scope is limiting. To fully unleash the power of AI, Team TransformCare recommends opening the following data through secure APIs: (1) social risk and SDOH data, such as ICD-10 Z-codes, which help AI agents understand and respond to patient needs, including transportation and food insecurity, (2) lab results and vital signs to enrich personal coaching with factual data pulled from claims/EHR, (3) medication fill data, including OTC usage to incorporate known over-the-counter medications of supplements for identifying potential adherence issues or interactions, and (4) provider information and appointment

data to expose upcoming appointments (with consent) so as to provide AI assistance with reminders, rescheduling, or finding open slots. (TD-2a, TD-2b, TD-2d)

CMS has an opportunity to fill data gaps by expanding Blue Button and making more data available (with privacy safeguards) to directly empower patients by providing them with the complete picture of their health.

5.1.3 Target Areas for Technology Solutions

CMS should direct emphasis toward objective, data-driven, technology-enhanced solutions that address three critical healthcare problems:

1. **Needless Care Variation** – Reducing unnecessary differences in treatment approaches for similar conditions
2. **Delivery of Low-Value Care** – Eliminating treatments that provide little to no benefit relative to their cost and risk
3. **Healthcare Inequities and Disparities** – Ensuring fair access to quality care regardless of demographics or geography

These systemic healthcare challenges have reached societal importance and underpin an estimated $1-1.25 trillion in wasteful healthcare spending annually in the U.S.

5.1.4 Technology Infrastructure Requirement

Solutions to these pervasive healthcare delivery issues will require novel knowledge management systems leveraging the latest AI platforms capable of:

- Advanced data representation architectures and management of multi-modal, personal, patient-level health data, information, and knowledge
- AI-enhanced operationalization of complex health information for real-world decision making
- Delivering information dominance – giving patients and providers complete, actionable health information
- Creating decision advantage – enabling better choices through comprehensive data analysis

These systemic healthcare challenges have reached societal importance and contribute significantly to an estimated $1-1.25 trillion in wasteful healthcare spending annually in the U.S.

5.1.5 Developer Incentive Framework

Short-term steps (following two years):

- Establish clear technical standards for SDM platforms
- Create pilot programs with Medicare Advantage plans
- Provide development grants for qualifying SDM technologies
- Streamline regulatory pathways for patient-centered decision tools

Longer-term steps:

- Integrate SDM requirements into Medicare reimbursement models
- Mandate SDM platform capabilities for certain high-impact clinical areas
- Scale successful pilot programs nationally
- Create sustainable funding mechanisms for ongoing platform development

5.1.6 Market Opportunity

The real stimulation for developer interest lies in solving large, societally important problems that deliver:

- Objective, observable outcomes that can be measured and validated
- Tangible life-altering benefits for health and wellness
- Clear return on investment through reduced waste and improved care
- Scalable solutions that can impact millions of Medicare beneficiaries

By focusing on these fundamental healthcare challenges, CMS can create a compelling market opportunity that attracts top-tier technology developers while directly benefiting Medicare beneficiaries and the broader healthcare system.

5.1.7 CMS Data and API Requirements

Additional Data Sources for Innovation

All available patient-level data is valuable for developing new knowledge management systems, including ontology-governed personal clinical knowledge graphs, machine deep learning models, reasoning engines, and sophisticated fine-tuned LLMs. These advanced systems require comprehensive data to function effectively and provide meaningful insights for patient care. (TD-2a)

Current Data Access Challenges

The primary challenges preventing access to valuable data sources today are status quo thinking, organizational inertia, lack of political will, and subtle but pervasive data blocking practices. These barriers prevent the healthcare system from realizing the full potential of available health information. (TD-2c)

5.1.8 Comprehensive API Strategy

CMS should prioritize developing all APIs required to serve the best interests of the Medicare beneficiary population. The patient must be the central focus of all decisions and initiatives. Both patients and providers benefit from truly operationalized, interoperable (structural and logical) patient-level health data.

Patients and providers benefit from time-relevant, context-aware "information dominance" and "decision advantage"—both achievable with new knowledge management systems. The current Electronic Health Information (EHI) data architecture is incapable of performing against these requirements. (TD-2d)

5.2 Digital Identity

(No responses for: TD-3a, 3b, 3c)

5.3 Technical Standards and Certification

(No responses for: TD-5 – 7b, TD-9b – 9e; TD-10 – 11c)

5.3.1 Open Standards vs Proprietary APIs

TransformCare agrees that standards-based APIs are preferable to proprietary systems. The unified AI-powered platform is built with interoperability at its core, the orchestration engine communicating via FHIR® APIs (R4) to retrieve or update health records. The platform also conforms to standards like HL7® v2 and C-CDA when exchanging information with hospital systems. All data our agents generate (call notes, patient-reported metrics) can be output in standardized formats so that it flows into existing IT systems. We encourage CMS and ONC to continue policies that reward use of public standards, for example: (TD-4, TD-9c)

- Advancing the requirement that certified EHRs support bulk FHIR data export (for population health)
- Expanding SMART on FHIR apps capabilities
- Providing incentives or penalties tied to open standard adoption
- Creating certification bonuses for systems that exceed minimum interoperability requirements

In our experience, some providers still resort to custom CSV or flat-file exchanges because certain data (e.g., free-text notes or scanned documents) is not exposed cleanly via API. Certification criteria could be tightened to ensure APIs cover "all aspects of the patient's chart, "including unstructured data, perhaps through emerging standards for text or image exchange.

By prioritizing open, standardized approaches over proprietary solutions, CMS can accelerate innovation while ensuring patient data portability and system interoperability.

5.3.2 Standards and Certification Requirements

Establishing a Digital Health Sandbox and voluntary "Safe Harbor" certification for non-diagnostic, safety-tested AI health agents is a valid alternative certification approach for CMS consideration. Rather than traditional FDA-style approvals, this approach would be a CMS-led validation program to review digital health products for efficacy, quality, and safety in improving outcomes. (TD-8, TD-9a)

Team TransformCare has demonstrated through a rigorous evaluation that generative AI agents can achieve safety parity with clinicians. In over 307,000 patient-agent interactions, our AI provided correct medical advice 99.38% of the time without making a single error that could have caused severe patient harm. By engaging over 6,200 licensed clinicians to test our AI, we eliminated major safety concerns and reached human-level accuracy. A CMS sandbox could allow developers to pilot similarly safety-optimized AI tools on Medicare use cases (e.g., post-discharge follow-ups, care navigation, preventative care education) under real-world conditions. High-performing solutions that emerge from the

sandbox would earn a “Safe Harbor” designation, giving providers and ACOs confidence to adopt them without fear of regulatory uncertainty.

The Safe Harbor certification concept provides a concrete example of how certification could be reimagined to better support innovation while maintaining safety standards.

5.4 Data Exchange

(No responses for: TD-12, TD-14 – 17)

The ASTP strategic framework provides clear direction for healthcare transformation. Their vision of "better health enabled by data" and mission "to create systemic improvements in health and care through the access, exchange, and use of data" requires fundamental changes to current systems. (TD-13)

ASTP's strategic goals include advancing person-centered and self-managed health, transforming healthcare delivery and community health, fostering research and scientific knowledge, and enhancing the nation's health IT infrastructure.

To realize this vision, mission, and strategic goals, ASTP will need to facilitate and support a substantially new data architecture platform capable of representing, visualizing, and analyzing scaled patient-level EHI. The current problems hampering these stated objectives are vastly deeper and more complex than simply choosing additional APIs.

From a “First Principles” perspective, both a new data representation architecture platform and a new knowledge management system will be required to unleash AI-based innovation that truly supports patients and providers.

5.5 Compliance

(No responses for: TD-18 – 19d)

6. Value-Based Care Organizations

Value-based care organizations represent the next evolution in healthcare delivery, where success depends on outcomes rather than volume. In a consumer-driven healthcare marketplace, these organizations thrive by competing on transparent quality metrics and cost efficiency while empowering patients to make informed choices. Marketplace-driven Alternative Payment Models (APMs) create the infrastructure for this transformation, aligning provider incentives with patient outcomes and enabling true competition based on value.

6.1 Digital Health Adoption and Knowledge Management Systems

6.1.1 Advanced Data Architecture for Value-Based Care

The Current Challenge

Digital health management must fundamentally start with optimal data representation, visualization, and operationalization. While databases (current EHRs) impose structural

constraints, they do not define logical rules or knowledge. Meaning and operational context are not inherent in legacy EHR platforms. Advanced knowledge frameworks, reasoning engines, and knowledge management systems must provide advanced machine-learning analysis, contextual meaning, and actionable healthcare insights. (VB-1)

The Transformative Solution

Access to the entirety of a patient's EHI provides the opportunity to completely reconfigure health data representation entirely. When a patient's EHI is fully represented by ontology-governed personal clinical knowledge graphs (KGs), the data becomes a structured, semantically rich form of "code," unlocking powerful capabilities. Knowledge graphs transform raw clinical data (labs, provider notes, test results, diagnoses) into standardized, interconnected entities and multidimensional relationships that enable machine learning models to perform predictive functions, time-sensitive reasoning, and pattern recognition.

Technical Architecture Requirements

Understanding the distinction between different data systems is crucial for APM success:

- Graph databases store relationships explicitly but do not infer new ones
- Knowledge bases contain structured knowledge with defined semantic frameworks and reasoning rules
- Knowledge management systems (KMS) integrate multiple sources, process updates dynamically, and enable user interaction through queries, reasoning, machine-learning analysis, and decision-support tools

Advanced Capabilities

Ontological governance enables symbolic AI reasoning and supports explainable decision-making. LLMs fine-tuned on medical ontologies can provide clinical summarization and natural language reasoning grounded in ontological structure. This advanced knowledge graph architecture enforces semantic data interoperability and facilitates comprehensive data operationalization through AI analysis, reasoning, inference, and querying capabilities.

The Outcome

The key requirement for realizing these capabilities is access to patient-level EHI data at sufficient scale to build paradigm-changing data representation models and knowledge management systems. This comprehensive approach ultimately produces "information dominance" and "decision advantage"—all in service to the patient. Without this foundational, comprehensive data access, the transformative potential of advanced health AI remains unrealized.

6.1.2 APM Data Operationalization Requirements

APMs must be able to operationalize patient-level and patient cohort-level data in a time-dependent, context-aware fashion to make better, faster, point-of-care, and forward-looking decisions at scale. Current commercial EHR data architectures do not support these necessary capabilities.

Optimal data operationalization fundamentally requires understanding structural versus logical interoperability and distinguishing between databases, knowledge bases, and knowledge management systems. This understanding is crucial for designing systems that efficiently and effectively drive information dominance and decision advantage at scale across populations.

To truly achieve health-data interoperability and knowledge-driven systems, there is an immediate need for semantic layers, reasoning engines, and new knowledge frameworks. When these capabilities are firmly in place, optimal data operationalization becomes an expected outcome—the ultimate incentive is **better, faster decisions at scale**. (VB-1)

6.1.3 Current Adoption Obstacles

The primary obstacles preventing broader digital product adoption for patients in ACOs are the inflexibility and isolation of current EHR systems. These systems create barriers that prevent the dynamic, interconnected approach necessary for effective value-based care. (VB-1a)

6.2 Marketplace-Driven APM Model

Digital health management must fundamentally start with optimal data representation, visualization, and operationalization. While databases (current EHRs) impose structural constraints, they do not define logical rules or knowledge. Meaning is not inherent in an EHR database — it must be provided by ontologies, reasoning engines, and knowledge management systems. Consider the difference between a graph database, a knowledge base, and a knowledge management system (KMS): (VB-1)

- A graph database stores relationships explicitly but does not infer new ones.
- A knowledge base contains ontology-guided structured knowledge with defined semantics and reasoning rules.
- A knowledge management system (KMS) integrates various sources, facilitates reasoning, and supports decision-making by capturing and processing knowledge dynamically.

In a knowledge management system (KMS):

- The system integrates multiple knowledge sources.
- It processes updates dynamically, ensuring that machine learning-based derived insights are maintained over time.
- Users interact with the system through queries, reasoning, and decision-support tools.

APMs must be able to operationalize patient-level and patient cohort-level data in a time-dependent, context-aware fashion to make better, faster, point-of-care, and forward-looking decisions at scale. Current commercial EHR data architectures do not support these necessary capabilities. Optimal data operationalization fundamentally requires understanding structural vs. logical interoperability and distinguishing between databases, knowledge bases, and knowledge management systems. This understanding is crucial for

designing systems that are efficient and effective at driving information dominance and decision advantage.

To truly achieve health-data interoperability and knowledge-driven systems, there is an immediate need for semantic layers (ontologies), reasoning engines, and new knowledge frameworks. When these new capabilities are firmly in place, optimal data operationalization can be an expected outcome.

Better, faster, smarter decisions at scale is the ultimate incentive.

The current obstacle to broader digital product adoption for patients in ACOs is the "inflexibility" and "isolation" of the current EHR systems.

6.2.1 AI-Powered Integration Requirements

The key technologies—artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement—can only be optimally leveraged and integrated into APM requirements when foundational patient-level data is functionally operationalized in a time-sensitive, context-aware, point-of-decision fashion.

Current commercial EHR data architectures do not support functional operationalization. To solve these problems, advanced knowledge frameworks are needed to provide the semantic blueprint for developing modern, AI-ready data representation platforms. When patient-level data is structured in this format, AI-augmented machine learning models and advanced fine-tuned LLM queries can be developed and leveraged. (VB-2)

6.2.2 Patient-Facing Integration

The key themes and technologies referenced can only be optimally leveraged and integrated into APM requirements when the foundational patient-level data, from which all decisions are derived, is functionally operationalized in a time-sensitive, context-aware, point-of-decision fashion. Current commercial EHR data architectures do not support functional operationalization. To solve these problems, an "ontology" is needed. The ontology provides the semantic (architectural) blueprint for the development of a modern-day, AI-ready, data representation platform—the personal, clinical knowledge graph. When patient-level data is structured in this format, AI-augmented machine learning models and advanced fine-tuned and prompt-augmented LLM queries can be developed and leveraged. (VB-2)

6.3 Technical Infrastructure Requirements

6.3.1 Essential Health IT Capabilities

Essential health IT capabilities for value-based care arrangements must include comprehensive data operationalization capabilities that support real-time decision making. These capabilities encompass care planning, patient event notification, data

extraction and normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, and other patient empowerment tools. (VB-3)

6.3.2 Platform Capabilities

To thrive as an APM, the marketplace requires robust health IT capabilities that unify data, enable informed choice, and facilitate transparent transactions. It needs a secure data aggregation layer that compiles longitudinal patient records from multiple sources, along with AI-driven recommendations to guide patients toward providers who meet their cost and quality preferences. Clear cost and quality metrics must be presented in an intuitive "consumer-like" interface so that patients can compare available services with ease. Digital identity and consent features are also critical, ensuring each party securely accesses only the data and functionalities they need. Finally, payment and risk-sharing mechanisms must be embedded into the platform, enabling the marketplace to function as a true value-based model that rewards providers demonstrating both efficiency and effectiveness. (VB-3)

6.3.3 Comprehensive Data Requirements

Successful participation in a marketplace-APM requires all available patient-level data across multiple domains: clinical data, claims and administrative data, patient-reported outcome data, social determinants of health data, care coordination and longitudinal data, quality measures and benchmarking data, cost and financial data, and real-time and remote monitoring biometric data. (VB-4)

For optimal VBC platform performance, there must be structural and logical data interoperability as well as sophisticated AI-enabled data analysis that supports risk stratification, care gap assessment, care coordination, and outcome performance and attribution. By amalgamating these comprehensive data types under standardized formats, the marketplace can deploy AI algorithms to match the right patient to the right service at the right time.

A Repeating Theme: On-Demand, AI-Enhanced Information Dominance and Decision Advantage

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6.3.4 Advanced Technology Requirements

Unlike typical APM participants who may still rely on traditional EHR workflows, organizations forming or joining a marketplace-based APM require real-time availability checks, cost visibility, and outcome analytics that link directly to patient decisions. They must deploy robust tools for AI-driven referrals and integrate pricing data from multiple payers or suppliers. By contrast, non-marketplace providers may not face the same competitive pressures to display costs or respond quickly to patient queries. Thus, a marketplace-APM participant's technology stack necessarily includes bidirectional

communication with patients, streamlined integration with claims systems, and tools for analyzing performance across the entire continuum of care. (VB-9)

6.4 Certification Program Enhancements

6.4.1 Current Baseline

The current ONC Health IT Certification Program criteria supply the baseline interoperability crucial for any marketplace-APM. Requirements such as FHIR-based APIs and electronic clinical quality measurement tools allow digital platforms to pull structured data from multiple EHRs. By leveraging these foundational standards, a marketplace can blend clinical data, cost details, and care outcomes into a single interface. However, current certification lacks explicit support for real-time "shopping" functionalities or transparent pricing modules that are central to a competitive healthcare marketplace. (VB-5)

6.4.2 Missing Capabilities

Several advanced capabilities remain unaddressed by current certification programs. Dynamic care coordination and cost-comparison features—enabling "one-click" booking and transparent pricing—are not systematically included. AI oversight and standardization, particularly regarding patient recommendations, risk stratification, and algorithmic explainability, remain in early stages. Remote monitoring and ePayment integrations, which help patients purchase services or supplies directly through the platform, also lack standardized certification pathways. Including these functionalities as optional but recognized modules under the ONC framework would fill crucial gaps for marketplace-APMs looking to transform healthcare from point-of-care transactions to longitudinal, value-based experiences. (VB-6)

6.4.3 Modular Approach

A marketplace-APM can reduce complexity by encouraging a more modular approach to certification and data exchange. Instead of mandating rigid, comprehensive requirements, regulators could specify core interoperability and security standards, then allow participants to adopt specialized modules—like AI-driven recommendations or telehealth billing—based on their unique patient populations or clinical focus. Measuring success predominantly through outcomes, rather than prescriptive checklists, provides necessary latitude for platforms to innovate in how services are priced, delivered, and tracked. (VB-7)

6.4.4 CMS Certification Recommendations

CMS could introduce certification criteria tailored to marketplace-based APMs, requiring advanced functionalities like unified billing gateways, dynamic pricing displays, and AI-enabled population health analytics. It could also endorse standards that ensure patients receive integrated cost and quality comparisons from a single platform. Remote monitoring and telehealth capabilities could be expanded to ensure seamless data flows into the marketplace so outcomes can be tracked in real time. Clear benchmarks for patient satisfaction, adherence rates, and provider response times would hold

marketplace participants accountable while enabling data-driven refinements of service offerings. (VB-10)

6.5 Implementation Support and Standards

6.5.1 Federal Support

HHS can further support the marketplace-APM model by introducing targeted incentives that reward price transparency, quality reporting, and timely service delivery. Funding pilot programs in underserved regions could demonstrate how digital marketplaces streamline care coordination, reduce unnecessary emergency visits, and improve access to specialty services. Clear guidance around privacy and cybersecurity (particularly for sensitive information like behavioral health data) could ease provider and patient concerns, encouraging broader participation. Additionally, forging partnerships with agencies like HRSA or SAMHSA would allow the marketplace concept to address unique local challenges, integrating community-based and social services into the digital ecosystem. (VB-8)

6.5.2 Overcoming Barriers

Common interoperability challenges include fragmented EHR ecosystems that do not natively support marketplace features, variable data quality across different provider organizations, and disparities in patients' digital literacy or internet access. These barriers hamper real-time matching of patients to services and undermine risk stratification and care coordination efforts. Privacy laws that differ by state further complicate the sharing of sensitive data, making it difficult to provide comprehensive service options in the marketplace platform. (VB-11)

6.5.3 Balanced Standards

A balanced approach would adopt core FHIR-based APIs and a universal digital identity standard (e.g., NIST 800-63 compliance) to authenticate users, patients, and providers across the marketplace. This core standardization ensures data liquidity without preventing APMs from tailoring their services to specific patient populations. The marketplace architecture should allow for specialized modules that build on universal interoperability rules while maintaining flexibility for different care models. (VB-12)

6.5.4 Enhanced Interoperability

Enhancing existing criteria and standards to better support value-based care requires a more holistic approach to interoperability and data exchange. For marketplace-based APMs, this means allowing standardized, near-real-time APIs that not only pull discrete clinical data but also integrate pricing, claims, and social determinants of health information under a common framework. Certification programs could move beyond static measures of data exchange toward ensuring that systems can handle dynamic interactions—like online scheduling, cost comparisons, and AI-driven recommendations—in one streamlined user experience. (VB-13)

6.6 Consumer Empowerment Infrastructure

6.6.1 Digital Identity Integration

Digital identity credentials can underpin the free-market concept by allowing patients and providers to seamlessly access the marketplace. Suppose patients can use a single credential (e.g., Login.gov, ID.me) to log into the consumer platform. In that case, they can compare service offerings, see real-time provider availability, and check coverage details without navigating multiple portals. Providers likewise benefit from easier onboarding across multiple healthcare networks or facilities, helping them participate in various marketplace-driven APMs without duplicating credentialing efforts. (VB-14)

6.6.2 Nationwide Provider Directory

A nationwide provider directory with FHIR endpoints would help the marketplace platform dynamically match patients with the most appropriate services. Patients could more easily identify services aligned with their specific health needs and budget constraints if the directory also contained data on provider specialties, cost structures, quality ratings, and telehealth availability. Essential elements include provider identifier (NPI), location, specialty, insurance network affiliation, scheduling availability, and pricing details. This "marketplace-ready" directory would accelerate both consumer choice and real-time care coordination, empowering patients and APM participants to optimize quality and cost outcomes. (VB-15)

6.7 Marketplace APMs as Free-Market Catalysts

Value-based care organizations operating within marketplace-driven APMs represent the practical implementation of free-market healthcare principles. By competing on transparent quality metrics and cost efficiency while empowering patient choice, these organizations demonstrate how market forces can drive healthcare transformation. When providers must compete for patient preference based on outcomes and value, the entire system becomes more responsive to actual patient needs rather than administrative convenience.

The result is a healthcare ecosystem where value-based care organizations thrive by serving patients better, creating sustainable competitive advantages through superior outcomes, transparent pricing, and exceptional patient experiences—the foundation of successful free-market healthcare.

7. CMS Leads or is Left Behind

The U.S. healthcare system continues to underperform despite spending $4.9 trillion annually—more than any other high-income country. Billions are wasted on administrative inefficiencies and antiquated payment models, while systemic complexities create opaque pricing, limit competition, and discourage individual responsibility.

7.1 The Solution is Within Reach

The advanced knowledge management systems and AI-powered platforms detailed throughout this response represent the pathway from our current failed system to the free-market, patient-centered transformation we outlined. Every principle we advocated—price competition, transparency, efficiency, and patient empowerment through data ownership—becomes achievable through these technologies.

7.2 The Harsh Reality for CMS

While Medicare beneficiaries—America's most vulnerable population—struggle with fragmented care and limited choices, private insurers and international health systems are deploying AI-powered platforms that fundamentally transform patient engagement, clinical decision-making, and care delivery. Every day CMS delays is another day 65 million beneficiaries are denied the empowered, transparent healthcare they deserve.

7.3 CMS has an Unprecedented Opportunity

With 65 million lives and massive purchasing power, CMS can drive the very transformation we described in our opening vision—putting patients at the center of value-based care with complete data ownership, transparent pricing, and genuine choice. By implementing TransformCare's platform, CMS transforms Medicare and sets the standard for American healthcare.

Every day of inaction leaves 65 million beneficiaries stranded in yesterday's healthcare system.

8. Team TransformCare

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Docsnap, Inc. is a digital consumer health utility headquartered in Atlanta, Georgia, founded in 2013 by Dr. Anthony Mari and Darrell Borne. The company empowers individuals to take control of their healthcare by providing seamless access to their complete medical records, regardless of provider or health system.

Docsnap addresses the challenge of healthcare data fragmentation by assigning each member a universal patient identifier. This enables users to request, aggregate, and standardize their health information into an actionable format accessible at the point of care.

Through its mobile platform, Docsnap offers features such as secure messaging with providers, 24/7 virtual care, and the ability to share records with family members and care teams. This approach enhances safety, reduces redundant procedures, and lowers healthcare costs for patients, providers, and payers alike.

Docsnap's mission is to revolutionize the healthcare experience by placing patients at the center of their care, ensuring timely access to vital health information, and fostering informed medical decisions.

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Recorded Health Inc. is a U.S.-based software company committed to developing platform-based solutions that empower patients, healthcare providers, and payers to consolidate, integrate, analyze, and collaborate around patient-level clinical, operational, and financial data. Our solutions leverage modern data representation architectures and AI-powered analytics to drive better, faster, and smarter decision-making. Data privacy, security, and regulatory compliance are foundational to our work. At our core, we believe that operationalizing data in a patient-centered, secure, and scalable manner is essential for meaningful change in healthcare delivery.

The U.S. healthcare system is in urgent need of a shift toward a model that prioritizes patient sovereignty, data control, and transparent data representation—ultimately flipping the existing power dynamic. We are honored to work toward these goals and to help catalyze a reimagined healthcare future. Recorded Health, along with our solution

partners, is pleased to provide our perspectives to inform CMS and ASTP/ONC in their efforts to assess, adopt, and deploy patient-empowerment technologies. We view this initiative as mission-critical—a necessary catalyst for a true U.S. healthcare reformation.

Our response to this RFI is based on experience gathered over decades of hands-on patient care, the development of technologies that measurably improve clinical, operational, and financial outcomes, and translational research that bridges innovation and real-world application. We hope our insights and recommendations contribute meaningfully to CMS and ASTP/ONC's mission to deliver ever-better, ever-improving health and well-being to the individuals and communities they serve.

Visit Recorded Health Inc. at www.recordedhealth.com for more information.

Hippocratic AI
— Do No Harm —

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Hippocratic AI is a safety-focused, HIPAA-compliant, patient-centric generative AI company dedicated to delivering "healthcare abundance"—making high-quality care infinitely scalable and accessible. We develop non-diagnostic AI healthcare agents that engage patients via natural conversations (e.g., phone, chat) for low-risk care navigation, follow-ups, education, and administrative support. Our platform emphasizes patient safety, prevention, and empowerment as core design principles.